Detecting and Recovering from Deception in Model-Free Nash-Equilibrium Seeking: A Dual-Demodulation Analysis

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Abstract

We justify a model-free deception detector for Nash-equilibrium (NE) seeking, built entirely from the demodulation signal C_v each player already computes. Three facts are proved: (i) *blindness* — C_v vanishes at the deceptive equilibrium exactly as at an honest NE, so it cannot certify honesty; (ii) *detection* — a second demodulation C'_v at a probe frequency the deceiver does not inject into makes $D_v = |C'_v - C_v|$ equal to the injection magnitude, giving a threshold that separates the two regimes; (iii) *recovery* — re-randomizing the victim's probe frequency collapses the deceptive equilibrium and restores exponential convergence to the true NE. We delimit all three. The detector is exactly evadable by a deceiver that mirrors its injection onto the second probe, at a cost of only $O(a^2)$ to the attack, so the guarantee is one of *probe secrecy* rather than of unconditional soundness; against a deceiver that respects secrecy, however, detectability and profit are locked together, and an attack that stays below the threshold delivers a gain that vanishes with the threshold. Recovery drives the deceiver's integral state out of any bounded set, so the post-switch dynamics terminate the deception in one of two ways — saturation, or unbounded and trivially visible injection — rather than leaving δ_d irrelevant. We further show that the detector's residual is *not* monotone in the probing amplitude a: it decomposes into a curvature bias $O(a^2)$ and a ripple bias $O\big((k/a\omega)^2\big)$ that *grows* as a shrinks, so the correct design variable is k/ω and there is an optimal amplitude $a^\star \propto \sqrt{k/\omega}$. All statements are validated on the true time-varying system: the identities hold to under 1.2%, the mirroring evasion drives D_v below the honest floor, the post-recovery windup rate matches $\varepsilon\gamma$ to 8%, the residual fit is $\eta \approx 0.19\,(k/a\omega_{\min})^2$ with 1.9% spread, and the profit–detectability frontier is monotone with correlation 0.9998.

1 Setup

Two players, a victim v and a deceiver d, have scalar actions $x_v, x_d \in \mathbb{R}$ and costs $J_v(x_v, x_d)$, $J_d(x_v, x_d)$ (vector / N-player case: Remark 14). Actions carry sinusoidal probes,

$$x_v = u_v + a\sin(\omega_v t)\ \big(+\ a\sin(\omega'_v t)\big), \tag{1}$$

$$x_d = u_d + a\sin(\omega_d t) + a\,\delta_d(t)\sin(\omega_v t), \tag{2}$$

with probing amplitude $a > 0$; the optional ω'_v-probe is used only by the detector of §4. Each player passes its measured payoff through a washout (high-pass) filter with corner $\omega_h > 0$ before demodulating,

$$\dot z_i = -\omega_h\big(z_i - J_i(x)\big), \qquad h_i := J_i(x) - z_i, \qquad i \in \{v, d\}, \tag{3}$$

and updates its slow state by demodulating the filtered payoff at its own frequency,

$$\dot{u}_v = -\tfrac{2k}{a}\, h_v \sin(\omega_v t), \qquad \dot{u}_d = -\tfrac{2k}{a}\, h_d \sin(\omega_d t), \qquad k > 0. \tag{4}$$

Note that the victim's *update* uses only ω_v; the second probe enters the action (1) but drives no update, a point that matters in §5. The deceiver injects the victim's probe with amplitude driven by integral feedback toward a target payoff J_d^{ref}:

$$\dot{\delta}_d = \varepsilon\big(J_d(x) - J_d^{\mathrm{ref}}\big), \qquad \varepsilon > 0. \tag{5}$$

Setting $\delta_d \equiv 0$ recovers honest NE seeking. Partials $\partial_v J_v := \partial J_v/\partial x_v$, $\partial_d J_v := \partial J_v/\partial x_d$, etc. are evaluated at the slow state $u = (u_v, u_d)$, and $\overline{f} := \lim_{T\to\infty} \frac{1}{T}\int_0^T f\, dt$ is taken with u, δ_d frozen.

Remark 1 (Why the washout is not optional). Without (3) the update (4) is driven by J_v itself, whose DC component $J_v(u)$ contributes a term $-\frac{2k}{a} J_v(u)\sin(\omega_v t)$; although this averages to zero, it integrates to a ripple in u_v of amplitude $2k|J_v(u)|/(a\omega_v)$. Unless $\omega_v \gg 2k|J_v|/a^2$ this ripple exceeds the probe amplitude a and the loop destabilises — an a^{-2} requirement that is impractical for small a. The washout removes $J_v(u)$ and leaves a signal of size $O(a)$, reducing the ripple to $O(k/\omega)$; see Remark 10. Filtered extremum seeking in this form is standard [2].

Assumption 1 (Regularity)**.** $J_v, J_d \in C^3$ on an open operating set $U \subset \mathbb{R}^2$ containing the trajectories of interest.

Assumption 2 (Probe and filter selection)**.** The frequencies $\omega_v, \omega_v', \omega_d > 0$ are distinct and non-resonant: $m_1\omega_v + m_2\omega_v' + m_3\omega_d \neq 0$ for all integers with $0 < |m_1| + |m_2| + |m_3| \leq 3$. The washout corner is placed between the slow and fast time scales, $k \ll \omega_h \ll \omega_{\min} := \min\{\omega_v, \omega_v', \omega_d\}$.

Assumption 2 is the usual orthogonality condition of extremum seeking; it yields

$$\overline{\sin(\omega_i t)\sin(\omega_j t)} = \tfrac{1}{2}\mathbb{1}[\omega_i = \omega_j], \qquad \overline{\sin(\omega_i t)\sin(\omega_j t)\sin(\omega_\ell t)} = 0. \tag{6}$$

The bound $|m| \leq 3$ is exactly what the leading-order argument needs; the third-order Taylor remainder generates combinations with $|m| = 4$ (for instance $3\omega_v$, from $\sin^3\omega_v t$), but those enter only the $O(a^2)$ bias and not the identity itself. Section 7 confirms both halves of this: hop targets violating $|m| \leq 3$ degrade recovery by up to a factor of 19, while a target at $3\omega_v$ — resonant only at $|m| = 4$ — is indistinguishable from a non-resonant one, in a quadratic and in a cubic game alike. The next assumption fixes the adversary. It is stated here, once, because it delimits *both* the detector of §4 and the recovery of §6; §5 shows that neither survives its removal.

Assumption 3 (Probe secrecy)**.** The deceiver knows ω_v and injects at that frequency only. It neither knows nor injects at any other frequency the victim uses — in particular not at the monitoring probe ω_v' of §4, nor at the post-switch frequency ω_v'' of §6.

Assumption 4 (Bounded injection)**.** The injection amplitude is bounded: there is $\bar{\delta} < \infty$ with $|\delta_d(t)| \leq \bar{\delta}$ for all t, and the corresponding trajectories remain in U. Equivalently, (5) carries a saturation or anti-windup element at level $\bar{\delta}$.

Assumption 4 is vacuous while deception succeeds — at a DNE the integrator (5) is at rest, so δ_d is automatically bounded — and it is *not* vacuous after recovery, which is precisely the content of Corollary 8. Constants written $O(a^2)$ below depend on $\bar{\delta}$ through the second- and third-order Taylor coefficients of J_v evaluated on the enlarged perturbation $a(1+\bar{\delta})$; this dependence is uniform on U by Assumption 1.

Averaging and the order of limits. We analyse the averaged system obtained below and record the probing bias as $O(a^2)$. Transfer to the true time-varying system is by standard averaging [3, Ch. 10], but the quantifiers matter and are stated here once, since they are the source of a design rule that is easy to get backwards: *for each fixed $a > 0$ there exists $\bar{\omega}(a) < \infty$ such that for probe frequencies exceeding $\bar{\omega}(a)$ the true trajectories remain $O(a)$-close to the averaged ones*, uniformly in t under the stability hypothesis of Theorem 7. The threshold $\bar{\omega}$ *depends on* a and grows as $a \downarrow 0$. Consequently one may not fix ω and then send $a \to 0$; doing so eventually violates the averaging hypothesis, and the resulting error is quantified in Remark 10.

2 Averaged dynamics and the deceptive equilibrium

Everything rests on the demodulation of the victim's filtered payoff at a frequency ω. Because the washout (3) has transfer function $s/(s+\omega_h)$, a component of J_v at frequency ω is attenuated by $\omega/\sqrt{\omega^2+\omega_h^2}$ and phase-shifted by $\arctan(\omega_h/\omega)$; demodulating with $\sin(\omega t)$ therefore retains the fraction

$$\Gamma(\omega) := \frac{\omega^2}{\omega^2+\omega_h^2} \in (0,1). \tag{7}$$

We define the *compensated* demodulation

$$\mathcal{D}_\omega[J_v] := \frac{2}{a\,\Gamma(\omega)}\,\overline{h_v \sin(\omega t)}. \tag{8}$$

Compensation uses only ω_h and the victim's own probe frequency, both of which it chooses, so (8) remains model-free. Up to the factor $\Gamma(\omega_v)$ this is exactly the quantity the victim already forms in (4): on average $\dot{u}_v = -k\,\Gamma(\omega_v)\,\mathcal{D}_{\omega_v}[J_v]$.

Lemma 1 (Demodulation returns a perceived gradient)**.** *Under Assumptions 1–2 and 4, with the deceiver injecting $a\delta_d \sin(\omega_v t)$:*

(a) $C_v := \mathcal{D}_{\omega_v}[J_v] = \partial_v J_v(u) + \delta_d\,\partial_d J_v(u) + O(a^2)$;

(b) if the victim also probes with amplitude a at $\omega_v' \notin \{\omega_v, \omega_d\}$ and Assumption 3 holds, then $C_v' := \mathcal{D}_{\omega_v'}[J_v] = \partial_v J_v(u) + O(a^2)$.

Proof. With u, δ_d frozen, a third-order Taylor expansion of J_v about u gives

$$J_v(x) = J_v(u) + \partial_v J_v\,(x_v - u_v) + \partial_d J_v\,(x_d - u_d) + R_2, \qquad R_2 = O(a^2),$$

where $x_v - u_v$ and $x_d - u_d$ are read off (1)–(2) and the implied constants are uniform in δ_d by Assumption 4. The washout is linear and annihilates the constant $J_v(u)$; each remaining sinusoidal component at frequency ω survives with gain $\Gamma(\omega)^{1/2}$ and phase $\arctan(\omega_h/\omega)$, so that after demodulation at that same frequency the net factor is $\Gamma(\omega)$, which (8) divides out. (a) Multiply h_v by $\frac{2}{a\Gamma(\omega_v)}\sin(\omega_v t)$ and average. By (6),

$$\tfrac{2}{a}\partial_v J_v\overline{(x_v-u_v)\sin(\omega_v t)} = \tfrac{2}{a}\partial_v J_v \cdot a\tfrac{1}{2} = \partial_v J_v,$$

$$\tfrac{2}{a}\partial_d J_v\overline{(x_d-u_d)\sin(\omega_v t)} = \tfrac{2}{a}\partial_d J_v\big(a\underbrace{\overline{\sin\omega_d t\,\sin\omega_v t}}_{=0} + a\delta_d\underbrace{\overline{\sin^2\omega_v t}}_{=1/2}\big) = \delta_d\,\partial_d J_v.$$

Each second-order term of R_2 is a product of two probe sinusoids times $\sin(\omega_v t)$, hence averages to zero by (6) regardless of its individual filter gain; the third-order remainder is $O(a^3)$ and contributes $\frac{2}{a}O(a^3) = O(a^2)$. (b) Multiply instead by $\frac{2}{a\Gamma(\omega_v')}\sin(\omega_v' t)$. The $\partial_v J_v$ term again yields $\partial_v J_v$ via the victim's own ω_v'-probe, while by Assumption 3 the deceiver injects nothing at ω_v', so $\overline{(x_d - u_d)\sin(\omega_v' t)} = 0$ because $\omega_v' \neq \omega_d, \omega_v$. The $\partial_d J_v$ contribution therefore drops. □

Remark 2 (Compensation is necessary, not cosmetic). If the raw demodulations $\frac{2}{a}\overline{h_v \sin(\omega t)}$ are used instead of (8), Lemma 1 returns $\Gamma(\omega_v)\big(\partial_v J_v + \delta_d \partial_d J_v\big)$ and $\Gamma(\omega_v')\partial_v J_v$. Since $\Gamma(\omega_v) \neq \Gamma(\omega_v')$, the true gradient no longer cancels in the difference and the detector of §4 acquires the spurious floor

$$\big|\Gamma(\omega_v') - \Gamma(\omega_v)\big|\,\big|\partial_v J_v\big| = \omega_h^2 \Big|\tfrac{1}{\omega_v^2} - \tfrac{1}{\omega_v'^2}\Big|\,\big|\partial_v J_v\big| + O(\omega_h^4),$$

which does not vanish with a and is largest exactly where the detector must be trusted — it is zero at an honest NE, where $\partial_v J_v = 0$, and grows with the honest gradient during the transients on which Remark 4 relies. In the configuration of §7 the formula gives a coefficient of 0.0327, so an honest transient with $|\partial_v J_v| \approx 0.2$ would carry a spurious floor of about 0.0067: the same size as the entire true residual η, and a fifth of the threshold ρ.

Lemma 1(a) is the mechanism of deception: $\dot{u}_v = -k\Gamma(\omega_v)C_v$ drives u_v to a zero of the *perceived* gradient $\partial_v J_v + \delta_d \partial_d J_v$, not of $\partial_v J_v$. Absorbing $\Gamma(\omega_v) = 1 + O(\omega_h^2/\omega_v^2)$ into k, the averaged closed loop is

$$\dot{u}_v = -k\big(\partial_v J_v(u) + \delta_d \partial_d J_v(u)\big) + O(a^2), \quad \dot{u}_d = -k\,\partial_d J_d(u) + O(a^2), \quad \dot{\delta}_d = \varepsilon\big(J_d(u) - J_d^{\mathrm{ref}}\big) + O(a^2). \tag{9}$$

Definition 1 (Honest and deceptive NE)**.** $u^\star$ is a true NE if $G_{\mathrm{hon}}(u^\star) = 0$, where $G_{\mathrm{hon}}(u) := \big(\partial_v J_v(u),\ \partial_d J_d(u)\big)$. A triple (u^*, δ_d^*) with $\delta_d^* \neq 0$ is a *deceptive Nash equilibrium* (DNE) if it is an equilibrium of (9):

$$\partial_v J_v(u^*) + \delta_d^* \partial_d J_v(u^*) = 0, \qquad \partial_d J_d(u^*) = 0, \qquad J_d(u^*) = J_d^{\mathrm{ref}}. \tag{10}$$

The first identity is the crux: $\partial_v J_v(u^*) = -\delta_d^* \partial_d J_v(u^*)$, so the victim's true gradient at the DNE is generically nonzero, of magnitude $|\delta_d^*||\partial_d J_v(u^*)|$. The third identity is what makes deception worth running — J_d^{ref} is chosen strictly better for the deceiver than the honest value $J_d(u^\star)$ — and it is what makes the integrator diverge once the victim escapes (Corollary 8).

3 Blindness of the raw signal

Lemma 2 (Single-frequency blindness)**.** *Along* (9)*,* $C_v = -\frac{1}{k}\dot{u}_v + O(a^2)$*. Hence* $C_v = O(a^2)$ *at every equilibrium of the victim's slow dynamics — at an honest NE* ($\delta_d = 0$*,* $\partial_v J_v(u^\star) = 0$*) and at a DNE* ($\delta_d = \delta_d^* \neq 0$*, perceived gradient zero by* (10)*) alike. No test on* C_v *alone can separate the two regimes.*

Proof. The identity is (9). In both cases the bracketed term vanishes, so $\dot{u}_v = O(a^2)$ and the two limits coincide. □

The same argument kills the action-based residual $r_v = x_v(t{+}1) - \hat{x}_v(t{+}1)$: the victim *does* follow the honest update rule, only its input J_v is corrupted, so $r_v = O(a^2)$ by construction. The defect is not the residual idea but the use of a single compromised frequency.

Remark 3 (Blindness is not merely asymptotic in the simulation). On the true system both limits sit far below the $O(a^2)$ floor they are only guaranteed to respect: §7 reports $|C_v| = 1.5 \times 10^{-13}$ honest against 6.1×10^{-5} deceptive, with $a^2 = 2.5 \times 10^{-3}$. The ratio between them is large, but the gap is an artifact of noiseless integration, not an exploitable signal: it survives no measurement noise at all. At $\sigma = 10^{-4}$ the two values are 4.4×10^{-5} and 6.2×10^{-5}, a ratio of 1.4, while D_v is unchanged at 0.0068 versus 0.2057. The action residual behaves the same way, 1.6×10^{-2} honest against 1.5×10^{-2} deceptive.

4 A detector from a second demodulation

Take a second copy of (8) at the monitoring frequency ω_v' and set

$$D_v := \left|C_v' - C_v\right|. \tag{11}$$

Theorem 3 (Detection identity). *Under Assumptions 1–4, along* (9),

$$D_v = |\delta_d| \left|\partial_d J_v(u)\right| + O(a^2) \qquad \textit{at every time } t. \tag{12}$$

Proof. Subtracting Lemma 1(a) from (b), $C_v' - C_v = -\delta_d \partial_d J_v(u) + O(a^2)$: the true gradient $\partial_v J_v(u)$ appears in both demodulations and cancels, leaving exactly the injection term that C_v could not isolate. Take absolute values. □

Remark 4 (Why the difference and not C_v' alone). At the DNE $C_v = O(a^2)$ by Lemma 2, so D_v and $|C_v'|$ agree *there*. They do not agree during honest transients, where $C_v' = C_v \neq 0$ and hence $D_v = O(a^2)$ while $|C_v'|$ is as large as the honest gradient. Testing $|C_v'|$ therefore raises an alarm on every honest run that has not yet converged; the subtraction is what buys a detector valid at all times rather than only after settling, which is the content of the phrase "at every time t" in (12). The cancellation is also what makes the detector sharper than either demodulation on its own: in §7 each of C_v, C_v' carries a bias of about 0.010, while their difference has a floor of 0.0068, because the two biases are largely common-mode.

Assumption 5 (Effective deception). There is $c_0 > 0$ with $|\partial_d J_v(u)| \geq c_0$ for all $u \in U$. Deception is *active* at time t if $|\delta_d(t)| \geq \delta_{\min}$ for a fixed $\delta_{\min} > 0$.

This only says deception is effective: if it were not, there would be nothing to detect.

Corollary 4 (Soundness, completeness, separation). *Let Assumptions 1–5 hold — in particular probe secrecy, Assumption 3 — and let*

$$\eta(a, k, \omega) := \sup_t \Big| D_v - |\delta_d| \left|\partial_d J_v(u)\right| \Big| \tag{13}$$

denote the total residual of (12) *on the true system. Suppose the design parameters are chosen so that*

$$\eta(a, k, \omega) \leq \tfrac{1}{4}\, \delta_{\min} c_0, \tag{14}$$

and fix $\rho := \frac{1}{2}\delta_{\min} c_0$. *Then under honest play* $D_v \leq \frac{1}{4}\delta_{\min} c_0 < \rho$ *for all* t; *whenever deception is active* $D_v \geq \delta_{\min} c_0 - \frac{1}{4}\delta_{\min} c_0 = \frac{3}{4}\delta_{\min} c_0 > \rho$. *Hence the rule "*$D_v > \rho \Rightarrow$ *deception,* $D_v \leq \rho \Rightarrow$ *honest" has neither false alarms nor missed detections* against a deceiver satisfying Assumption 3.

Proof. Honest play sets $\delta_d \equiv 0$ in (12), leaving $D_v \leq \eta$ within the bound (14). Active deception gives $|\delta_d||\partial_d J_v| \geq \delta_{\min} c_0$ by Assumption 5; subtracting η yields the lower bound. The ranges $[0, \rho)$ and (ρ, ∞) are disjoint. □

Corollary 4 decides a fixed attack. The next statement says something stronger about the attacker's design problem: the threshold does not merely classify, it prices. Write $\pi := J_d(u^\star) - J_d(u^*)$ for the deceiver's gain over honest play.

Proposition 5 (No quiet profit)**.** *Let Assumptions 1–6 hold (the last of these is stated in §6, where it is also used), and let $\delta \mapsto u^*(\delta)$ be the branch of solutions of the first two equations of (10) through $u^*(0) = u^\star$, which exists and is C^1 near $\delta = 0$ by the implicit function theorem, since the Jacobian there is $DG_{\mathrm{hon}}(u^\star)$, nonsingular by Assumption 6. Along this family, generated by varying J_d^{ref}:*

(i) $D_v \geq c_0|\delta| + O(a^2)$ at the corresponding DNE;

(ii) the gain $\pi(\delta) = J_d(u^\star) - J_d(u^*(\delta))$ is continuous with $\pi(0) = 0$.

Consequently, for every threshold $\rho > 0$ the gain available to a deceiver that the detector reads as honest is at most

$$\bar{\pi}(\rho) := \sup\big\{|\pi(\delta)| \ : \ c_0|\delta| \leq \rho + O(a^2)\big\}, \qquad \bar{\pi}(\rho) \longrightarrow 0 \quad as \quad \rho \to 0.$$

Proof. (i) is Theorem 3 together with Assumption 5. For (ii), at $\delta = 0$ the first two equations of (10) reduce to $G_{\mathrm{hon}}(u) = 0$, whose solution on the branch is $u^\star$ by Definition 1; continuity of $u^*(\cdot)$ and of J_d (Assumption 1) gives $\pi(0) = 0$ and continuity of π. The final claim follows because $D_v \leq \rho$ forces $c_0|\delta| \leq \rho + O(a^2)$ by (i), and $\bar{\pi}(\rho) \to \pi(0) = 0$ by continuity. □

Remark 5 (Greed is loud). Proposition 5 is qualitative — it gives no rate — but the mechanism is transparent and the numerics of §7 make it quantitative for a specific game: sweeping J_d^{ref} traces a frontier on which profit and D_v are monotonically related with correlation 0.9998, and the quietest attack that clears the threshold $\rho = 0.03$ recovers only 4.5% of the honest payoff. The statement is subject to Proposition 6: it constrains deceivers that respect probe secrecy, and says nothing about one that mirrors.

Remark 6. D_v uses only the victim's own payoff samples and its own probes at ω_v, ω_v' — no model of J_v, no knowledge of the deceiver's identity, of δ_d, or of J_d. It is admissible in the same model-free setting as the NE-seeking dynamics themselves. The constant c_0 enters only the calibration of ρ, which may equivalently be set from the empirical level of D_v during a known-honest interval.

5 The detector's attack surface

Corollary 4 is a statement about a deceiver that injects on one frequency. That restriction is not cosmetic: dropping it defeats the detector exactly, at negligible cost to the attack.

Proposition 6 (Exact evasion by a mirroring deceiver)**.** *Suppose the deceiver injects on both of the victim's probe frequencies,*

$$x_d = u_d + a\sin(\omega_d t) + a\,\delta_d(t)\sin(\omega_v t) + a\,\delta'(t)\sin(\omega_v' t),$$

in violation of Assumption 3. Then

$$D_v = \big|\delta' - \delta_d\big|\,\big|\partial_d J_v(u)\big| + O(a^2). \tag{15}$$

In particular the choice $\delta' \equiv \delta_d$ gives $D_v = O(a^2)$ for all t: the detector reads honest while deception is active with arbitrary strength. Moreover this evasion costs the deceiver nothing at first order — the victim's update (4) demodulates at ω_v only, so the added component averages out of $\dot{u}_v$ and the DNE (10) is unchanged to $O(a^2)$; the deceiver's own payoff acquires only the $O(a^2)$ probing bias of the extra perturbation.

Proof. Repeat Lemma 1(b) with the extra component present. Demodulating at ω_v', the term $\frac{2}{a}\partial_d J_v \cdot a\delta'\overline{\sin^2 \omega_v' t} = \delta'\partial_d J_v$ now survives, so $C_v' = \partial_v J_v + \delta'\partial_d J_v + O(a^2)$, while C_v is unchanged. Subtracting gives (15). For the cost claim, demodulating the added component at ω_v gives $\overline{\sin(\omega_v' t)\sin(\omega_v t)} = 0$ by (6), so it does not enter the averaged victim dynamics (9); it enters J_d only through second-order Taylor terms, contributing $O(a^2)$ to (5). □

Proposition 6 locates the guarantee precisely. Detection here is not an information-theoretic impossibility result for the deceiver; it is a *secrecy* statement, and its strength is exactly the strength of the victim's ability to keep ω_v' unknown. Three consequences are worth recording.

Remark 7 (Secrecy is a timing condition, not a static one). The deceiver's payoff J_d depends on x_v through $\partial_v J_d \neq 0$ — which is precisely what makes deception profitable in the first place — so a deceiver that spectrally scans its own payoff can in principle recover ω_v' after an acquisition time T_{acq} determined by its measurement noise and the probe amplitude a. Assumption 3 is therefore realistic only if the victim re-draws ω_v' on a timescale T_{hop} with

$$\frac{1}{\omega_{\min}} \ll T \ll T_{\text{hop}} < T_{\text{acq}},$$

where T is the detection window of Remark 11. The middle inequality says the detector must complete a decision between re-draws; the right one says the deceiver must not finish acquiring first. The right-hand side is not addressed here; the left-hand side is, in Remark 8.

Remark 8 (Hopping is free from a stability standpoint). Remark 7 turns the victim into a switched system, whereas Theorem 7 analyses a single switch, so one might expect a dwell-time condition to appear and to conflict with the requirement $T_{\text{hop}} < T_{\text{acq}}$. Numerically it does not. In §7 the victim cycles through five admissible frequencies at dwell times spanning 0.05–20 s — the shortest being a third of the washout time constant $\omega_h^{-1} = 0.159$ s and 1/67 of the slow time constant $k^{-1} = 3.33$ s, and amounting to 3600 switches in a single run — and the closed loop remains stable throughout, with recovery error growing only from 1.5×10^{-3} to 1.9×10^{-2}, still comfortably inside the $O(a) = 0.05$ band the theorem promises. This is a negative result, and a useful one: it removes an objection rather than adding a hypothesis. The victim's slow dynamics average over the switching, so the binding constraint on T_{hop} is the detection window, not stability.

Remark 9 (Toward a randomized detector). A natural strengthening is to give the victim a bank of m monitoring probes and to draw the one driving (4) at random. Blinding the resulting $\binom{m}{2}$ pairwise discrepancies requires the deceiver to mirror its injection across the whole bank, which multiplies its action perturbation by m and so is no longer free once the action is amplitude-constrained; mirroring only a subset leaves at least one pair with $\delta' \neq \delta_d$ and hence, by (15), a nonzero D_v. This converts the deterministic guarantee of Corollary 4 into a detection probability with an explicit security parameter m, at the cost of m additional non-resonant frequencies — a scarce resource in N-player games, where Assumption 2 must hold across all players simultaneously, and one whose scarcity is quantified by the resonance experiment of §7. We leave the analysis to future work.

Remark 10 (The residual is not monotone in a; design rule). It is tempting to read (12) as saying that (14) is met by taking a small. This is false on the true system, and the error is in the wrong direction. Two mechanisms contribute to η.

(i) *Curvature bias*, from the third-order Taylor remainder in Lemma 1: $O(a^2)$, decreasing in a.

(ii) *Ripple bias.* In (4) the prefactor $2k/a$ multiplies a filtered payoff of size $h_v = O(a)$, so $\dot{u}_v = O(k)$ *independently of* a; its oscillatory part, at the mixed frequencies $\omega_i \pm \omega_j$, integrates

to a ripple in u_v of amplitude $O(k/\omega_{\min})$, again independent of a. What matters for the demodulation is the ripple *relative* to the probe,

$$\nu := \frac{\text{ripple}}{a} = O\Big(\frac{k}{a\,\omega_{\min}}\Big),$$

which *grows* as $a \downarrow 0$. Its leading contribution to η is $O(\nu^2)$.

Hence

$$\eta \;\lesssim\; C_1 a^2 + C_2\Big(\frac{k}{a\,\omega_{\min}}\Big)^2, \tag{16}$$

which is minimised not at $a \to 0$ but at

$$a^\star = \Big(\frac{C_2}{C_1}\Big)^{1/4}\sqrt{\frac{k}{\omega_{\min}}}, \qquad \eta_{\min} = 2\sqrt{C_1 C_2}\,\frac{k}{\omega_{\min}}. \tag{17}$$

The design condition (14) is therefore a condition on $k/\omega_{\min}$, not on a alone:

$$\frac{k}{\omega_{\min}} \;\lesssim\; \frac{\delta_{\min} c_0}{8\sqrt{C_1 C_2}}.$$

This is consistent with, and is the quantitative form of, the order-of-limits caveat in §1: shrinking a at fixed ω eventually violates the averaging hypothesis. Section 7 measures $C_2 \approx 0.191$ with 1.9% spread across a sweep of a, k and an overall frequency scale, and finds $C_1 a^2$ below the resolution of the sweep throughout — so over the range examined the residual is governed by the ripple term alone, and a should be increased, not decreased, until the curvature term becomes visible.

Remark 11 (Finite averaging window). An implementation replaces $\bar{\cdot}$ by a sliding window $\frac{1}{T}\int_{t-T}^{t}$. This adds to η a window ripple $O(1/(\omega_{\min}T))$, and it requires u to be approximately frozen over the window if the identity is to hold through the transient, i.e. $T \ll 1/k$. The two requirements pull in opposite directions, but they are not symmetric in their consequences, and the interval between them is not automatically usable. *The lower bound is hard.* If the window ripple exceeds the intrinsic floor (16), it dominates η and produces false alarms outright. Since (14) must hold with the ripple included, admissibility requires $1/(\omega_{\min}T) \lesssim \delta_{\min}c_0$ at some T compatible with the upper bound — which is once again a condition on $k/\omega_{\min}$, not a consequence of time-scale separation as such. In the game of §7, $k \ll \omega_{\min}$ holds comfortably ($k/\omega_{\min} = 9.5 \times 10^{-3}$) and the nominal interval $(\omega_{\min}^{-1}, k^{-1}) = (0.03, 3.3)$ s is two decades wide, yet *every* window in it produces false alarms; the smallest usable window is $T = 10\text{ s} = 3k^{-1}$. *The upper bound is soft.* Exceeding $T \ll 1/k$ does not break the steady-state test; it degrades the transient identity error, and gradually. In §7 that error is smallest at $T = 10$ s and has merely doubled at $T = 40$ s. The practical rule is therefore to take the smallest T whose ripple sits below the floor of Remark 10, and to read $T \ll 1/k$ as a caution about transient fidelity rather than as a constraint. Identity (12) additionally requires δ_d to be slow relative to the window, $\varepsilon \ll 1/T$, or the two demodulations are compared at effectively different injection amplitudes.

6 Recovery: collapsing the deceptive equilibrium

Detection and recovery are two uses of the same primitive. At a trigger time t_0 (when $D_v > \rho$) the victim re-randomizes the frequency driving (4) from ω_v to a secret $\omega_v'' \notin \{\omega_v, \omega_d\}$ satisfying Assumptions 2 and 3.

Assumption 6 (Stable true NE)**.** The honest pseudogradient flow $\dot{u} = -kG_{\text{hon}}(u)$ has an equilibrium $u^\star$ (Definition 1) at which $-k\,DG_{\text{hon}}(u^\star)$ is Hurwitz (e.g. strongly monotone games).

Theorem 7 (Recovery)**.** *Let Assumptions 1, 2, 3, 4, 6 hold. There exists $\bar{a} > 0$, depending on $\bar{\delta}$, such that* for every $a \in (0, \bar{a})$ *there exist $\bar{\omega}(a) < \infty$ and $\bar{\varepsilon}(a) > 0$ with the following property: if the probe frequencies exceed $\bar{\omega}(a)$ and $\varepsilon \in (0, \bar{\varepsilon}(a))$, then after the switch $\omega_v \mapsto \omega_v''$ at t_0 the slow state $u(t)$ converges exponentially to an $O(a)$-neighborhood of $u^\star$, for every trajectory of $\delta_d(t)$* admissible under Assumption 4*; in particular $x_v(t) \to u_v^\star + a\sin(\omega_v'' t) + O(a)$.*

Proof. The victim now demodulates and updates at ω_v'' while the deceiver, by Assumption 3, still injects at ω_v. Repeating Lemma 1(a) with ω_v'' in the update and the injection left at ω_v, the cross term carries $\overline{\sin(\omega_v t)\sin(\omega_v'' t)} = 0$ by (6). The perceived gradient thus loses its bias and $\dot{u} = -kG_{\text{hon}}(u) + O(a^2)$; crucially δ_d no longer appears in the u-dynamics, so the deceiver's integral state (5) is decoupled from u. The decoupling alone does not make δ_d irrelevant, because the $O(a^2)$ term is uniform in δ_d only on bounded sets: the neglected Taylor remainders carry factors $a^2\delta_d^2\,\partial_{dd}^2 J_v$ and $a^3\delta_d^3\,\partial^3 J_v$, so an unbounded δ_d would destroy both the perturbation bound and the containment of x_d in U required by Assumption 1. Assumption 4 supplies the needed uniformity: on $|\delta_d| \le \bar{\delta}$ the remainder is bounded by $C(\bar{\delta})a^2$ with C independent of t. By Assumption 6, $u^\star$ is locally exponentially stable for $\dot{u} = -kG_{\text{hon}}(u)$; with a quadratic Lyapunov function V satisfying $\dot{V} \le -c\|u - u^\star\|^2$ near $u^\star$, the $C(\bar{\delta})a^2$ probing bias is a vanishing perturbation, so for $a < \bar{a}(\bar{\delta})$ the averaged system converges exponentially to an $O(a^2)$-ball around $u^\star$. By the averaging theorem [3, Thm. 10.4], this exponential stability transfers to the true time-varying system up to an $O(a)$-neighborhood once the probe frequencies exceed a threshold $\bar{\omega}(a)$ determined by the chosen a; the small-ε condition places the decoupled integrator (5) in standard singular-perturbation form [3, Ch. 11] and does not affect the u-limit. □

Remark 12 (The switch target must be admissible, and this is not a formality)*.* Theorem 7 requires ω_v'' to satisfy Assumption 2 against the frequencies already in use, so "re-randomize" presupposes a characterised support set — a point easy to pass over, since the proof needs only $\overline{\sin\omega_v t \sin\omega_v'' t} = 0$, which any distinct frequency provides. The higher-order terms are what bite. In §7, hopping to $\omega_v'' = 2\omega_v$ (resonant via $m = (2, -1, 0)$) or to $\omega_v'' = \omega_v + \omega_d$ (via $m = (1, -1, 1)$) leaves recovery errors 4× and 19× larger than an admissible target; both violate Assumption 2 exactly as stated, at $|m| = 3$. Conversely $\omega_v'' = 3\omega_v$, which is resonant only at $|m| = 4$ and therefore permitted by Assumption 2, performs as well as any admissible target — in a quadratic game, where $\partial^3 J \equiv 0$, and also in a cubic one where it does not. The bound $|m| \le 3$ is thus tight in both directions: weakening it admits targets that cost an order of magnitude, and strengthening it to $|m| \le 4$ would exclude targets that cost nothing.

Assumption 4 is not a technical convenience: without it the post-switch dynamics do not merely fail to satisfy the hypotheses of Theorem 7, they actively violate them, and in a way that is itself informative.

Corollary 8 (Post-recovery windup)**.** *Suppose the deceiver runs (5) without saturation and the conclusion of Theorem 7 holds on $[t_0, t_1]$, so that $J_d(u(t)) \to J_d(u^\star)$. Assume the deception was worth running, $\gamma := J_d(u^\star) - J_d^{\text{ref}} \neq 0$. Then for every $\theta \in (0, 1)$ there is t_1 with*

$$\big|\delta_d(t)\big| \;\ge\; \big|\delta_d(t_1)\big| + \theta\,\varepsilon\,|\gamma|\,(t - t_1), \qquad t \ge t_1,$$

so δ_d leaves every bounded set in finite time and the deceiver's action perturbation $a|\delta_d|$ grows without bound.

Proof. By Theorem 7, $u(t) \to u^\star$ up to $O(a)$, hence $J_d(u(t)) - J_d^{\mathrm{ref}} \to \gamma$ by continuity (Assumption 1); choose t_1 so that $|J_d(u(t)) - J_d^{\mathrm{ref}}| \geq \theta|\gamma|$ with fixed sign for $t \geq t_1$. Integrating (5) gives the bound. The conclusion of Theorem 7 is unaffected by the growth of δ_d over $[t_1, t]$ only as long as x_d remains in U; the corollary asserts that this eventually fails. □

Remark 13 (Recovery terminates the deception either way). Corollary 8 closes the loop that Theorem 7 leaves open. A deceiver whose integrator is unsaturated cannot simply persist after the victim hops: its injection amplitude diverges linearly at rate $\varepsilon|\gamma|$, and an injection of unbounded amplitude at a known frequency is detectable by any spectral monitor, quite apart from the actuation limits it violates. A deceiver that does saturate at $\bar{\delta}$ satisfies Assumption 4, and Theorem 7 applies verbatim: the victim converges to $u^\star$ while the injection persists uselessly at a frequency no one demodulates. A deceiver that abandons the attack sets $\delta_d \to 0$ and the game is honest. All three are exhibited in §7, and the victim's final error is the same to within 2×10^{-4} in all of them: the deception ends in every case, and the only difference is whether the deceiver's own state stays bounded. The asymmetry the deceiver exploited — a single influence channel at a frequency the victim was obliged to keep using — is gone the moment the victim is free to move.

Remark 14 (Vector and N-player extension). For $x_v \in \mathbb{R}^{n_v}$ and several deceivers, replace $\partial_v J_v, \partial_d J_v$ by the Jacobian blocks $\nabla_v J_v, \nabla_d J_v$, the scalar δ_d by the injection-amplitude vector δ, and absolute values by Euclidean norms. Lemma 1 holds componentwise, (12) becomes $D_v = \|(\nabla_d J_v)\delta\| + O(a^2)$, Assumption 5 becomes the coercivity bound $\|(\nabla_d J_v)\delta\| \geq c_0\|\delta\|$ on U, Assumption 4 becomes $\|\delta\| \leq \bar{\delta}$, and Assumption 6 becomes Hurwitzness of $-k\, DG_{\mathrm{hon}}(u^\star)$. Theorem 3, Corollary 4, Proposition 5, Theorem 7 and Corollary 8 carry over; Proposition 6 becomes $D_v = \|(\nabla_d J_v)(\delta' - \delta)\| + O(a^2)$, so evasion requires the mirroring condition $(\nabla_d J_v)(\delta' - \delta) = 0$, which for $n_v < n_d$ admits solutions with $\delta' \neq \delta$ — the deceiver need only match the injection modulo $\ker(\nabla_d J_v)$.

7 Numerical validation

Every run below integrates the *true* time-varying system (1)–(5) by RK4, never the averaged one; the averaged system satisfies the identities by construction and would prove nothing. The base game is the strongly monotone quadratic

$$J_v = \tfrac{1}{2}(x_v - 1)^2 + c\, x_v x_d + e\, x_d, \qquad J_d = \tfrac{1}{2}(x_d - 1)^2 + c\, x_v x_d, \qquad c = 0.3,\ e = 0.5,$$

with $a = 0.05$, $k = 0.30$, $\varepsilon = 0.08$, probe frequencies $2\pi\{5, 7.3, 11.7\}$ for $\{\omega_v, \omega_d, \omega_v'\}$, washout $\omega_h = 2\pi$, $J_d^{\mathrm{ref}} = 0.15$, step $\Delta t = 5 \times 10^{-4}$ (about 27 steps per period of the fastest probe), sliding window $T = 10$, threshold $\rho = 0.03$. A cubic variant, $J_v + \frac{q}{3}x_v^3$ with $q = 0.4$, is used in §7 (h) to load the third-order remainder, which vanishes identically in a quadratic game.

(a) Analytic ground truth. Every "predicted" entry below comes from one place: a simultaneous solve of the three conditions (10), performed before any simulation. This fixes a single value per quantity and removes the possibility of a number being fitted after the fact.

quantity	value
true NE $u^\star$	$(0.769231,\ 0.769231)$
honest deceiver payoff $J_d(u^\star)$	0.204142
DNE u^*	$(0.544467,\ 0.836660)$
injection δ_d^*	0.308342
victim's true gradient $\partial_v J_v(u^*)$	-0.204535 (not zero)
coupling $\partial_d J_v(u^*)$	0.663340
predicted $D_v = \lvert\delta_d^*\rvert\lvert\partial_d J_v(u^*)\rvert$	0.204535
windup rate $\varepsilon\gamma$, $\gamma = J_d(u^\star) - J_d^{\text{ref}} = 0.054142$	4.331×10^{-3}

The three conditions (10) are satisfied at this point to 1.7×10^{-14}, 2.8×10^{-17} and 1.6×10^{-15} respectively. Note that the fifth and seventh rows coincide: that is identity ($\star$), not two independent measurements.

(b) The DNE exists where the theory says. Honest and deceptive runs, averaged over $t \in (340, 400)$:

quantity	analytic	simulated	|diff|
honest u_v	0.76923	0.76118	8.1×10^{-3}
deceptive u_v	0.54447	0.54238	2.1×10^{-3}
deceptive u_d	0.83666	0.83841	1.8×10^{-3}
δ_d^*	0.30834	0.30203	6.3×10^{-3}

All discrepancies are within the $O(a) = 0.05$ band the averaging theorem guarantees, and the two DNE coordinates are within $O(a^2) = 2.5 \times 10^{-3}$. The deception does what it is supposed to: the victim's cost rises from 0.5922 to 0.6603 while the deceiver's falls from 0.2023 to 0.1495, against a target of 0.150.

(c) Blindness, and the two demodulations separately. $|C_v|$ at steady state is 1.5×10^{-13} honest against 6.1×10^{-5} deceptive — both orders of magnitude below the $O(a^2) = 2.5 \times 10^{-3}$ floor they are only guaranteed to respect, so the apparent ratio between them carries no information; Remark 3 shows it collapses to 1.4 under noise of $\sigma = 10^{-4}$, at which D_v still separates the regimes by a factor of 30. The action residual is likewise blind, 1.6×10^{-2} against 1.5×10^{-2}. Lemma 1 holds with residuals $|C_v - (\partial_v J_v + \delta_d \partial_d J_v)| = 0.0114$ and $|C_v' - \partial_v J_v| = 0.0098$ in the deceptive run, and 0.0115, 0.0106 in the honest one — for scale, $|\partial_v J_v(u^*)| = 0.2045$.

(d) The detector. D_v measures 0.20571 against the analytic 0.20454, a relative error of 0.57%. The identity (12) holds along the trajectory, not only in the limit: the pointwise error against $|\delta_d(t)||\partial_d J_v(u(t))|$ averages 0.0035 over the transient $t \in (40, 240)$ and 0.0056 at steady state. The honest floor is $\eta = 0.00683$ (mean and max coincide to five figures) against a deceptive minimum of 0.20519, so the threshold $\rho = 0.03$ separates the regimes with a margin of 30× and neither a false alarm nor a miss occurs. Carrying the second probe displaces the DNE by 5.4×10^{-6}, three orders of magnitude inside a^2. Note that η is smaller than either individual demodulation bias of §7 (c), which is the common-mode cancellation of Remark 4.

(e) Exact evasion (Proposition 6). The deceiver injects $a\,\delta' \sin(\omega_v' t)$ with $\delta' = \mu\,\delta_d$:

mirror gain μ	0	0.25	0.5	0.75	1	1.25	1.5	2
D_v	0.2057	0.1559	0.1060	0.0561	0.0061	0.0440	0.0943	0.1952
DNE u_v	0.5424	0.5423	0.5421	0.5419	0.5417	0.5415	0.5412	0.5406
verdict		DECEPTION			HONEST		DECEPTION	

At $\mu = 1$ the detector reads 0.0061, *below* the honest floor 0.00683, while the attack proceeds at full strength; the DNE has moved by 6.6×10^{-4}, i.e. the evasion is free to within $O(a^2)$. The predicted V-shape $D_v = |\mu - 1|\,\delta_d |\partial_d J_v(u^*)|$ has slope 0.200 in these units; the measured branches give 0.1996 (left, 0.4%) and 0.1891 (right, 6.5%). The largest deviation of any of the eight points from the V is 0.0070, equal to η, and it changes sign across the vertex — consistent with the residual being a signed bias of that size rather than a noise floor, since $D_v = \left|(\delta' - \delta_d)\partial_d J_v + b\right|$ shifts by $\pm b$ according to the sign of the leading term.

(f) Recovery, windup, and the three exits (Theorem 7, Corollary 8). The victim switches to $\omega_v'' = 2\pi \cdot 13.9$ at $t_0 = 300$ and the run continues to $t = 1000$. The victim's action moves from 0.5460 (DNE 0.5445) to 0.7714, an error of 2.2×10^{-3} against the true NE 0.7692 — well inside $O(a)$. The deceiver's integrator, unsaturated, runs from 0.2969 at the switch to 3.5291 at $t = 1000$; the fitted growth rate is 4.68×10^{-3} against the predicted $\varepsilon\gamma = 4.33 \times 10^{-3}$, an error of 8.1%. At actuation limits $a|\delta_d| \le 0.5, 1, 2$ this gives times-to-visibility of roughly 2.4×10^3, 4.5×10^3 and 8.8×10^3 time units after the switch. The three exits of Remark 13:

deceiver behaviour	final u_v	error to $u^\star$	final δ_d
unsaturated	0.7714	2.15×10^{-3}	3.5291
saturated at $\bar{\delta} = 0.5$	0.7712	2.01×10^{-3}	0.5000
abandons ($\varepsilon = 0$)	0.7712	2.00×10^{-3}	0

(g) Hopping: dwell time and admissible targets. Cycling through five admissible frequencies with a fixed dwell (Remark 8) gives final errors 1.9×10^{-2}, 4.2×10^{-3}, 3.1×10^{-3}, 2.0×10^{-3}, 1.5×10^{-3} at dwells 0.05, 0.2, 1, 5, 20 s, with no divergence anywhere — the shortest dwell producing 3600 switches in one run, at a third of the washout time constant $\omega_h^{-1} = 0.159$ s. Accuracy degrades gracefully; stability does not degrade at all. For the choice of target (Remark 12), with $\{\omega_v, \omega_d\}/2\pi = \{5, 7.3\}$ fixed:

$\omega_v''/2\pi$	smallest $\lvert m\rvert \le 3$ combination	resonant?	error to $u^\star$
16.4	1.80	no	0.0014
$15.0 = 3\omega_v$	0.40	only at $\lvert m\rvert = 4$	0.0016
13.9	0.70	no	0.0021
$10.0 = 2\omega_v$	0, at $m = (2, -1, 0)$	yes	0.0059
$12.3 = \omega_v + \omega_d$	0, at $m = (1, -1, 1)$	yes	0.0267

Assumption 2 as stated separates the table exactly: the two targets it forbids are the two worst, by factors of 4 and 19, and the one it permits despite a $|m| = 4$ resonance is indistinguishable from the best.

(h) Residual scaling, window, noise, and a non-quadratic game. Sweeping $a \in \{0.025, 0.05, 0.10, 0.15\}$, $k \in \{0.1, 0.3\}$ and an overall frequency scale in $\{1, 2, 4\}$, the honest floor fits

$$\eta \;\approx\; 0.191 \left(\frac{k}{a\,\omega_{\min}}\right)^2,$$

with 1.9% spread across the eight configurations and no residual dependence on a, k or ω separately — so $C_2 \approx 0.19$ and $C_1 a^2$ is below the resolution of the sweep. Halving a from 0.05 to 0.025 *raises* η from 0.0068 to 0.0274, within 10% of the threshold $\rho = 0.03$: the naive rule "take a small" is backwards, as Remark 10 predicts. For the window (Remark 11), the honest maximum and the deceptive minimum over $T \in \{0.5, 1, 2, 3, 5, 10, 20, 40\}$ are $(0.231, 0.023)$, $(0.161, 0.068)$, $(0.082, 0.128)$, $(0.032, 0.182)$, $(0.032, 0.180)$, $(0.0068, 0.205)$, $(0.0068, 0.205)$, $(0.0068, 0.205)$. Every $T \le 5$ produces false alarms and $T = 0.5$ additionally produces misses, although $1/k = 3.3$ s; the transient identity error is smallest at $T = 10$ (0.0035) and rises to 0.0082 at $T = 40$. This is the asymmetry described in Remark 11. Additive Gaussian noise on the payoff readings leaves the test intact up to $\sigma = 0.02$ (honest 99th percentile 0.0227, deceptive 1st percentile 0.1898, no false alarms or misses at ρ); at $\sigma = 0.05$ the false-alarm rate is 25.7% with misses still at zero. The detector variance scales as $1/(a^2 T)$, in the same direction as the ripple bias, so noise strengthens rather than weakens the design rule $a^\star \propto \sqrt{k/\omega_{\min}}$. Finally, on the cubic game — where $\partial^3_{vvv} J_v = 0.8 \neq 0$ and the $O(a^3) \to O(a^2)$ step of Lemma 1 is genuinely loaded — D_v measures 0.18712 against the analytic 0.18922, a relative error of 1.11%, with an honest floor of 0.00628 and separation intact.

(i) The profit–detectability frontier (Proposition 5). Sweeping J_d^{ref} over the attainable range traces the constraint the attacker cannot escape while respecting Assumption 3:

J_d^{ref}	0.204	0.195	0.180	0.165	0.150	0.135	0.120
gain π	0.0001	0.0091	0.0241	0.0391	0.0541	0.0691	0.0841
δ_d^*	0.0008	0.0498	0.1333	0.2195	0.3083	0.4002	0.4952
D_v	0.0026	0.0378	0.0954	0.1516	0.2065	0.2602	0.3127
detected at $\rho = 0.03$?	no	yes	yes	yes	yes	yes	yes

Gain and detectability are monotonically related with correlation 0.9998. The quietest attack that clears the threshold recovers $\pi = 0.0091$, which is 4.5% of the honest payoff; anything quieter is, by Proposition 5, worth proportionately less. Lowering ρ tightens the admissible gain continuously, which is the content of $\bar{\pi}(\rho) \to 0$.

Summary of agreement. Every identity in the paper is confirmed on the true system: Lemma 2 (both regimes blind to within noise), Lemma 1 (residuals ≈ 0.01 against a signal of 0.20), Theorem 3 (0.57%, and 1.11% on the cubic game), Proposition 6 (V-shape to within η; detector defeated at $\mu = 1$), Theorem 7 (recovery error 2.2×10^{-3}), Corollary 8 (windup rate 8.1%), and Remark 10 ($\eta \approx 0.19\nu^2$, 1.9% spread). Two results correct the theory as originally stated rather than confirming it: Remark 11, where the nominal window interval turns out to be entirely unusable in a game satisfying every stated hypothesis, and Remark 8, where an anticipated dwell-time constraint does not exist.

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